

Markscheme

Mathematics: analysis and approaches
Practice question bank

45. (a)

Given that $\log_5 2 = 0.4307$ and $\log_5 3 = 0.6826$, both correct to 4 decimal places.

Find the value of $\log_5 720$, correct to 3 decimal places.

$$720 = 2^4 \times 3^2 \times 5 \quad (M1)$$

$$\log_5 720 = 4\log_5 2 + 2\log_5 3 + 1 \quad (M1)$$

$$= 4(0.4307) + 2(0.6826) + 1 \quad A1$$

$$= 4.088 \quad A1$$

[4 marks]